

MASENO UNIVERSITY
PRE-EXAMINATION REVISION PAPER

**YEAR I SEMESTER II EXAMINATION FOR MASTER OF SCIENCE IN
APPLIED STATISTICS**

MAS 812: NONPARAMETRIC METHODS

Time: 3 Hours

INSTRUCTIONS TO CANDIDATES:

1. Answer ANY THREE Questions.
2. Start answering each new question on a fresh page in the answer booklet.
3. Do not write anywhere on this question paper.
4. Statistical tables are provided at the end of this paper.
5. Observe further instructions on the answer booklet.

QUESTION ONE**[20 Marks]**

- a) Define the **empirical cumulative distribution function (ECDF)** $\hat{F}_n(x)$ and state three important properties of the ECDF. (4 marks)
- b) State the **Glivenko-Cantelli theorem** and explain its significance in nonparametric statistics. (2 marks)
- c) Let $X_1, X_2, \dots, X_n$ be i.i.d. from a continuous distribution with CDF F . For a fixed x , derive the mean and variance of $\hat{F}_n(x)$. Show that $\hat{F}_n(x)$ is an unbiased estimator of $F(x)$. (3 marks)
- d) State three key assumptions of parametric methods and explain how nonparametric methods relax these assumptions. (4 marks)
- e) A researcher collects the following dataset representing the body mass index (BMI) measurements of participants in a health study:

22.5, 24.1, 28.3, 26.7, 23.8, 31.2, 29.5, 25.4, 27.9, 33.6, 24.8, 30.1, 26.3, 21.9, 34.2

- i) Compute the **order statistics** for this dataset. Show your working. (1 mark)
- ii) Find the **median**, **first quartile** (Q_1), and **third quartile** (Q_3) using the order statistics. (3 marks)
- iii) Calculate the **interquartile range (IQR)** and identify any potential outliers using the $1.5 \times \text{IQR}$ rule. (3 marks)

QUESTION TWO**[20 Marks]**

- a) Define the **sign test** statistic T_+ for testing $H_0 : M = M_0$ versus $H_1 : M < M_0$, where M is the population median. State its exact distribution under the null hypothesis. (2 marks)
- b) Define the **Wilcoxon signed-rank test** statistic W^+ . State the assumptions required for this test and write down its mean and variance under the null hypothesis (assuming no ties). (5 marks)
- c) A researcher wants to test whether the median waiting time (in minutes) at a clinic is less than 20 minutes. A sample of 12 patients yields the following waiting times:

18, 22, 15, 19, 25, 16, 21, 14, 17, 23, 24, 13

- i) Perform the **sign test** at $\alpha = 0.05$. State your hypotheses clearly and show all steps. (6 marks)

- ii) Perform the **Wilcoxon signed-rank test** at $\alpha = 0.05$. Show all steps including ranking and handling of ties. (6 marks)
- iii) Compare the results of the two tests. Which test is more appropriate for these data and why? (1 mark)

QUESTION THREE**[20 Marks]**

- a) Define the following test statistics and state the null hypothesis each tests: (3 marks)
 - i) Wilcoxon rank-sum statistic W
 - ii) Median test statistic (Mood's median test)
 - iii) Kolmogorov-Smirnov two-sample statistic $D_{m,n}$

- b) A study compares the effectiveness of two different diets on weight loss (in kg). The results are:

Diet A: 4.2, 5.1, 3.8, 6.0, 4.5, 5.5, 4.9, 3.5 Diet B: 2.5, 3.2, 2.8, 4.0, 3.0, 3.5, 2.2, 2.9, 3.1

- i) Perform a **Wilcoxon rank-sum test** at $\alpha = 0.05$ to determine if the diets differ in weight loss. Show all steps including ranking and handling of ties. Compute both W and U . (8 marks)
- ii) Perform a **median test** (Mood's median test) at $\alpha = 0.05$ for the same data. Show all steps including construction of the contingency table and calculation of the chi-square statistic. (5 marks)
- iii) Perform a **Kolmogorov-Smirnov two-sample test** at $\alpha = 0.05$. Compute the test statistic $D_{m,n}$ and compare it to the critical value. (4 marks)

QUESTION FOUR**[20 Marks]**

- a) Define the following test statistics and state the design for which each is appropriate:
 - i) Kruskal-Wallis test statistic H
 - ii) Friedman test statistic F_r (2 marks)
- b) State the asymptotic distribution of the Kruskal-Wallis test statistic under the null hypothesis. Write down the formula for the tie correction factor and explain when it is needed. (2 marks)
- c) A researcher compares the effectiveness of three different teaching methods on test scores. Students are randomly assigned to one of the three methods, and their scores

are recorded:

Method A: 78, 82, 75, 80, 77, 81, 79, 76

Method B: 68, 72, 65, 70, 67, 71, 69, 66

Method C: 85, 88, 82, 87, 84, 86, 83, 81

- i) Perform a **Kruskal-Wallis test** at $\alpha = 0.05$ to determine if the teaching methods differ. (9 marks)
- ii) Conduct **post-hoc pairwise comparisons** using Dunn's test with Bonferroni correction and identify which methods differ significantly. (7 marks)

QUESTION FIVE

[20 Marks]

- a) A study measures the reaction times (in milliseconds) for 6 subjects under three different lighting conditions in a repeated measures design:

Subject	Dim Light	Normal Light	Bright Light
1	250	235	220
2	265	248	232
3	258	240	225
4	272	255	238
5	248	232	218
6	260	245	228

- i) Perform a **Friedman test** at $\alpha = 0.05$ to determine if the lighting conditions affect reaction time. Show all steps including within-block ranking. (6 marks)
 - ii) How would you conduct post-hoc comparisons using Wilcoxon signed-rank tests with an appropriate multiple comparison correction? (2 marks)
- b) Derive the computational formula for Spearman's rank correlation coefficient:

$$\rho = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)}$$

where d_i are the differences between the ranks of paired observations. State any assumptions. (6 marks)

- c) A study examines the relationship between daily coffee consumption (cups per day) and hours of sleep per night for 10 individuals:

Individual	1	2	3	4	5	6	7	8	9	10
Coffee	3	1	4	2	5	3	6	2	4	1
Sleep	7.5	8.2	6.8	7.9	6.2	7.4	5.8	8.0	6.5	7.8

- i) Compute **Spearman's rank correlation coefficient** ρ for these data. Show all steps including ranking and calculation of d_i . (4 marks)
- ii) Test whether ρ is significantly different from zero at $\alpha = 0.05$. State your hypotheses clearly. (2 marks)

STATISTICAL TABLES

Table 1: Critical Values for the Standard Normal Distribution

$z_{0.10}$	$z_{0.05}$	$z_{0.025}$	$z_{0.01}$	$z_{0.005}$	$z_{0.001}$
1.282	1.645	1.960	2.326	2.576	3.090

Table 2: Critical Values for the Chi-Square Distribution

df	$\chi^2_{0.10}$	$\chi^2_{0.05}$	$\chi^2_{0.025}$	$\chi^2_{0.01}$	$\chi^2_{0.001}$
1	2.706	3.841	5.024	6.635	10.828
2	4.605	5.991	7.378	9.210	13.816
3	6.251	7.815	9.348	11.345	16.266
4	7.779	9.488	11.143	13.277	18.467
5	9.236	11.070	12.833	15.086	20.515
6	10.645	12.592	14.449	16.812	22.458
7	12.017	14.067	16.013	18.475	24.322
8	13.362	15.507	17.535	20.090	26.124
9	14.684	16.919	19.023	21.666	27.877
10	15.987	18.307	20.483	23.209	29.588

Table 3: Critical Values for the Kolmogorov-Smirnov Goodness-of-Fit Test (One-Sample)

n	5	6	7	8	9	10	12	15	20	25	30
$D_{n,0.10}$	0.509	0.468	0.436	0.410	0.387	0.369	0.338	0.304	0.265	0.238	0.218
$D_{n,0.05}$	0.565	0.521	0.486	0.457	0.432	0.410	0.375	0.338	0.294	0.264	0.242
$D_{n,0.01}$	0.669	0.618	0.577	0.543	0.514	0.490	0.449	0.404	0.352	0.316	0.290

Table 4: Critical Values for the Kolmogorov-Smirnov Two-Sample Test ($\alpha = 0.05$)

m	n									
	5	6	7	8	9	10	12	15	20	30
5	0.800	0.767	0.743	0.725	0.711	0.700	0.683	0.667	0.650	0.633
6	0.767	0.733	0.714	0.700	0.689	0.680	0.667	0.650	0.633	0.617
7	0.743	0.714	0.690	0.679	0.670	0.662	0.650	0.633	0.617	0.600
8	0.725	0.700	0.679	0.662	0.654	0.646	0.633	0.617	0.600	0.583
9	0.711	0.689	0.670	0.654	0.641	0.633	0.617	0.600	0.583	0.567
10	0.700	0.680	0.662	0.646	0.633	0.620	0.600	0.583	0.567	0.550
12	0.683	0.667	0.650	0.633	0.617	0.600	0.575	0.558	0.542	0.525
15	0.667	0.650	0.633	0.617	0.600	0.583	0.558	0.540	0.520	0.500
20	0.650	0.633	0.617	0.600	0.583	0.567	0.542	0.520	0.500	0.480
30	0.633	0.617	0.600	0.583	0.567	0.550	0.525	0.500	0.480	0.460

Table 5: Critical Values for Spearman's Rank Correlation Coefficient

n	One-Sided α			Two-Sided α		
	0.05	0.025	0.01	0.10	0.05	0.02
5	0.900	0.900	1.000	0.900	1.000	1.000
6	0.829	0.886	0.943	0.771	0.886	0.943
7	0.714	0.786	0.893	0.679	0.786	0.893
8	0.643	0.738	0.833	0.619	0.738	0.833
9	0.600	0.700	0.783	0.583	0.700	0.783
10	0.564	0.648	0.745	0.549	0.648	0.745
11	0.536	0.618	0.709	0.521	0.618	0.709
12	0.503	0.587	0.678	0.491	0.587	0.678
13	0.484	0.560	0.648	0.473	0.560	0.648
14	0.464	0.538	0.622	0.454	0.538	0.622
15	0.446	0.521	0.604	0.438	0.521	0.604
16	0.429	0.503	0.585	0.421	0.503	0.585
17	0.414	0.485	0.566	0.406	0.485	0.566
18	0.401	0.472	0.550	0.394	0.472	0.550
19	0.391	0.460	0.535	0.384	0.460	0.535
20	0.380	0.447	0.522	0.374	0.447	0.522

Table 6: Critical Values for Kendall's Tau

n	One-Sided α			Two-Sided α		
	0.05	0.025	0.01	0.10	0.05	0.02
5	0.800	0.800	1.000	0.800	1.000	1.000
6	0.733	0.867	0.867	0.600	0.733	0.867
7	0.619	0.714	0.810	0.524	0.619	0.810
8	0.571	0.643	0.714	0.500	0.571	0.714
9	0.500	0.556	0.667	0.444	0.500	0.667
10	0.467	0.511	0.600	0.422	0.467	0.600
11	0.418	0.491	0.564	0.382	0.418	0.564
12	0.394	0.455	0.545	0.364	0.394	0.545
13	0.359	0.436	0.513	0.333	0.359	0.513
14	0.363	0.407	0.473	0.330	0.363	0.473
15	0.333	0.390	0.467	0.305	0.333	0.467
16	0.317	0.367	0.433	0.300	0.317	0.433
17	0.309	0.353	0.426	0.279	0.309	0.426
18	0.294	0.340	0.412	0.268	0.294	0.412
19	0.281	0.322	0.392	0.257	0.281	0.392
20	0.268	0.316	0.379	0.247	0.268	0.379

Table 7: Critical Values for the Wilcoxon Signed-Rank Test

n	Two-Sided α				
	0.10	0.05	0.02	0.01	0.001
5	1	1	-	-	-
6	2	2	2	-	-
7	4	4	3	2	-
8	6	6	5	4	2
9	8	8	7	6	4
10	11	11	9	8	6
11	14	14	12	11	8
12	17	17	15	14	10
13	21	21	19	17	13
14	26	26	23	21	16
15	30	30	27	25	20
16	36	36	32	30	24
17	41	41	38	35	28
18	47	47	43	41	33
19	54	54	50	47	38
20	60	60	56	53	43

Table 8: *

Note: Reject H_0 if $W^+ \leq$ critical value.

Table 9: Critical Values for the Wilcoxon Rank-Sum Test (Two-Sided $\alpha = 0.05$)

m	n									
	3	4	5	6	7	8	9	10	11	12
3	6,15	6,18	7,20	7,23	8,25	8,28	9,30	9,33	10,35	10,38
4	6,18	7,21	8,24	9,27	10,30	11,33	12,36	13,39	14,42	15,45
5	7,20	8,24	9,28	11,31	12,35	13,39	15,42	16,46	17,50	19,53
6	7,23	9,27	11,31	12,36	14,40	15,45	17,49	19,53	21,57	22,62
7	8,25	10,30	12,35	14,40	16,45	18,50	20,55	22,60	24,65	26,70
8	8,28	11,33	13,39	15,45	18,50	20,56	23,61	25,67	28,72	30,78
9	9,30	12,36	15,42	17,49	20,55	23,61	26,67	29,73	32,79	35,85
10	9,33	13,39	16,46	19,53	22,60	25,67	29,73	32,80	36,86	39,93
11	10,35	14,42	17,50	21,57	24,65	28,72	32,79	36,86	40,93	44,100
12	10,38	15,45	19,53	22,62	26,70	30,78	35,85	39,93	44,100	48,108

Table 10: *

Note: Entries are (lower critical value, upper critical value). Reject H_0 if $W \leq$ lower or $W \geq$ upper.

Table 11: Critical Values for the Kruskal-Wallis Test

df	Significance Level α				
	0.10	0.05	0.025	0.01	0.001
2	4.605	5.991	7.378	9.210	13.816
3	6.251	7.815	9.348	11.345	16.266
4	7.779	9.488	11.143	13.277	18.467
5	9.236	11.070	12.833	15.086	20.515
6	10.645	12.592	14.449	16.812	22.458

Table 12: Critical Values for the Friedman Test

k (treatments)	n (blocks)	Significance Level α			
		0.10	0.05	0.025	0.01
3	3	6.000	6.000	-	-
3	4	6.000	6.500	8.000	8.000
3	5	5.200	6.400	7.600	8.400
3	6	5.333	7.000	8.333	9.000
3	7	5.429	7.143	8.357	9.143
3	8	5.250	6.250	7.750	8.750
3	9	5.111	6.222	7.444	8.667
3	10	5.200	6.200	7.600	8.600
4	2	6.000	6.000	-	-
4	3	6.600	7.400	9.000	9.600
4	4	6.300	7.800	8.700	9.900
4	5	6.360	7.680	9.000	10.080
4	6	6.400	7.600	8.800	10.000